

QUANTUM CODES

When data is encrypted, only a person with the key can unscramble and read it. Quantum cryptography uses quantum phenomena such as superposition (see pp.38–39) or entanglement (see pp.72–73) to encrypt and decrypt data. This can make it impossible to snoop on encrypted conversations; an eavesdropper must measure a quantum key to gain information about it, causing the wave functions of the shared quantum keys to collapse, thus revealing the unauthorized access.

Secret codes

Alice shares a secret message with Bob using a series of polarized photons. Alice and Bob then agree on a “sifted key,” which is based on measurements made with compatible polarization filters.

